Problem Set 5: Question 1

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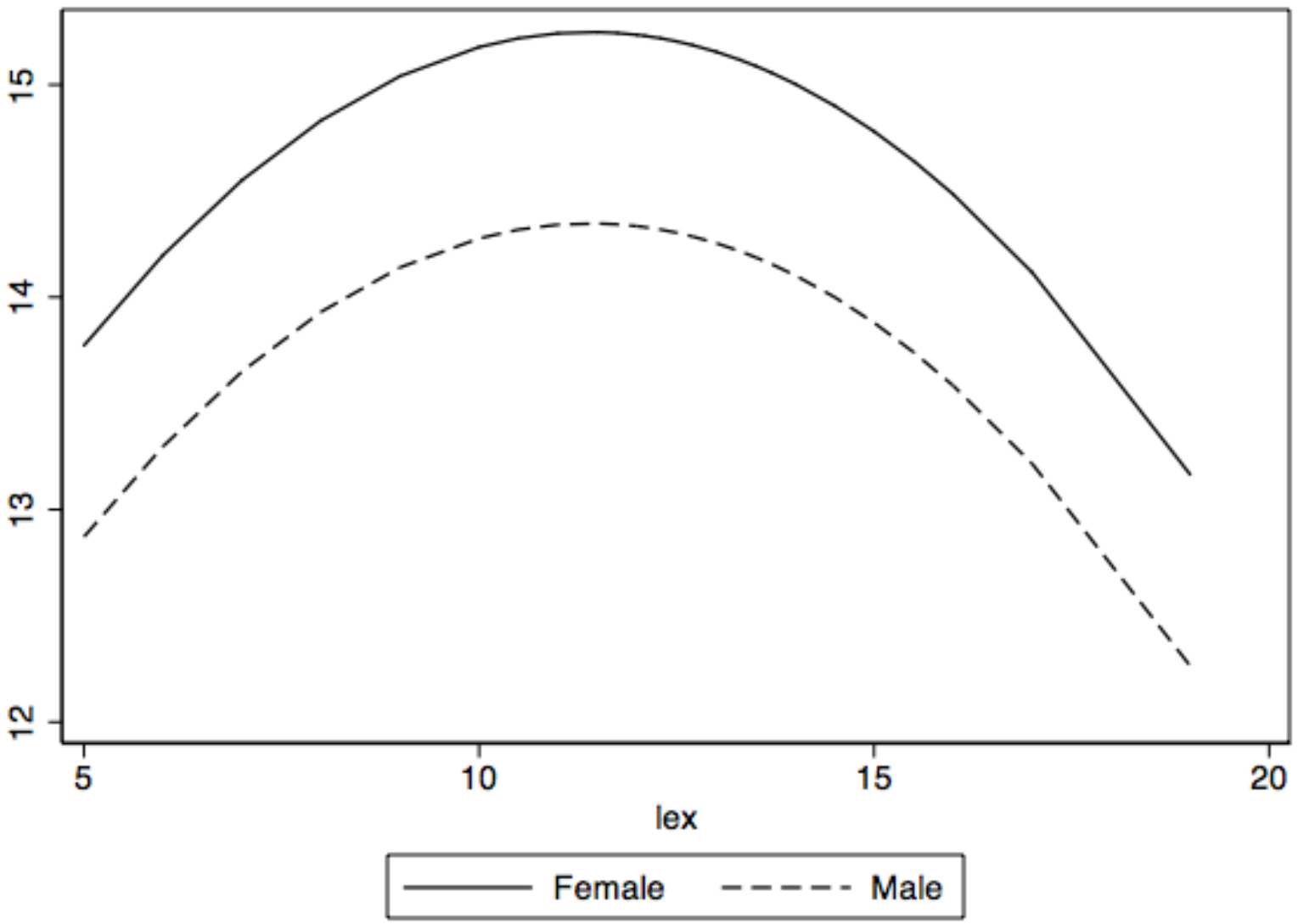
Estimation

- Estimate the folloing equation:
  $y = \alpha + \beta dex + \gamma_1 lex + \gamma_2 lex^2 + \theta sex + u$

OLS

	(1)
Dep var:	y
Sex	-0.901
	[0.087]***
Dex.	0.112
	[0.006]***
Education	0.821
	[0.321]**
Education^2	-0.036
	[0.013]***
Observations	683

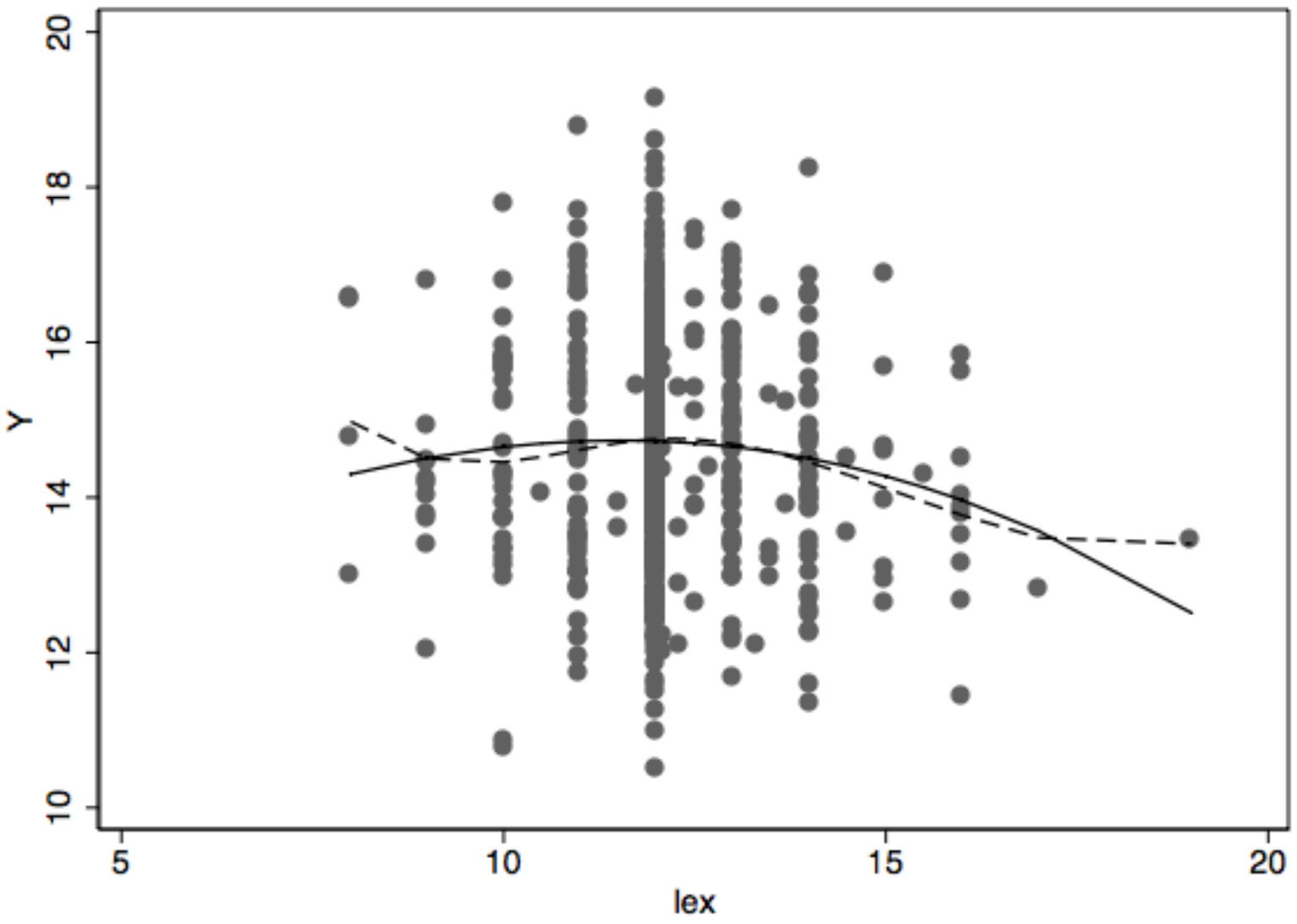
Effect of education



Effect of education by gender

OLS	
	(1)
Dep var	Y
Sex(=1)	-4.668
	[4.306]
Dex.	0.112
	[0.006]***
Education	0.486
	[0.522]
Education^2	-0.022
	[0.022]
Education*Sex(=1)	0.606
	[0.694]
Education^2*Sex(=1)	-0.024
	[0.028]
Observations	683

Maximizing productivity

	lex*	s.e.	95% CI
Delta Method	11.4	0.64	[10.14; 12.65]
Bootstrap	11.4	2.91	[5.70; 17.09]

Quadratic vs. cubic B-spline Education